

Upcoming Hackathons Ideas

FOSS - <https://fossunited.org/fosshack/rules> **(1st March to 31st March)**

Idea1: - Observed in SVIET too (Category - College Specific Challenges)

Many colleges currently collect admission documents through WhatsApp and manual file sharing. Admission staff often scroll through long chat histories to find student documents and receipts, leading to inefficiency and data mismanagement. We propose building an open-source, self-hostable web application that streamlines the admission process by centralizing document uploads, verification, and tracking.

The platform will allow admission cells to upload, organize, and verify student documents in a structured manner. AI-powered document verification (via OCR APIs) can be integrated to automatically extract and validate student information from uploaded certificates.

This system eliminates dependency on chat-based document exchange and provides a secure, searchable, and auditable admission workflow.

Some Enhancements:

Bulk Admission Mode (We built this in our company)

Upload Excel sheet of students + their documents in bulk → auto-create profiles.

Smart Missing Document Detection

System automatically shows:

- “Migration certificate missing”
- “Photo blurred”
- “Signature not detected”

QR Code Based Document Tracking

Generate a QR code for each student file. - Can just print this QR code to know the student all information - if all files are uploaded or not ?

Idea2: Open Source Document Intelligence Toolkit (Can be sub part of above tool)

Instead of only building admission software, build a reusable FOSS tool:

“DocVerify OSS”

A plug-and-play microservice that:

- Extracts text using OCR
- Detects tampered PDFs
- Compares name/DOB across documents

- Classifies document type (marksheet, Aadhaar, PAN, etc.)

PixelVerse: All about design + Case study of the design

Hack-N-Win:

Open theme: Women safety

Create an app which is not just like an existing SOS app.

Divide Escalation to 3 levels

Level 1: Notify trusted friends or volunteers (can register on app to help others)

Level 2: Notify campus security

Level 3: Direct police integration

“Safe Walk” Feature

- Student selects destination inside campus
- Nearby verified volunteers/security get notification
- Live tracking till safe arrival

Voice Alert Trigger system

Fake call simulation

Trigger fake call from “Mom” or “Security”

Exit unsafe social situations

Auto record 30 seconds to 1 minute (configurable) when SOS triggered.

TechNova:

Open Innovation: Women safety

WitchHunt: Can use FOSS project under Education category

Girls Safety in Colleges - Deepfake & Image Misuse.

AI-Based Deepfake & Image Misuse Detection System for Women’s Safety in Colleges

Problem:

With the rise of AI tools, female students in colleges are increasingly becoming victims of:

- Deepfake image/video morphing
- Non-consensual image editing

- Fake social media accounts using their photos
- Online harassment using manipulated media

Victims often:

- Don't know if the media is manipulated
- Don't know how to collect proof
- Feel scared to report
- Face mental trauma and reputational damage

Colleges lack:

- A rapid verification system
- A secure reporting mechanism
- Evidence-backed complaint tools

Proposed System:

“SHE-SHIELD” – AI Media Protection & Rapid Response Platform

A **college-integrated AI platform** that:

1. Detects deepfake or morphed content
2. Generates authenticity reports
3. Enables anonymous complaint submission
4. Alerts college Internal Complaints Committee (ICC)
5. Provides emotional & legal assistance resources

How It Works (System Design)

1. Deepfake Detection Engine

When a student uploads suspicious content:

- Run image forensic analysis:
 - Face warping detection
 - GAN fingerprint detection
 - Inconsistent lighting/shadow analysis

- Metadata analysis

Use:

- CNN-based deepfake classifier
- Pretrained models like EfficientNet
- Face landmark detection

Output:

- Authenticity score (0–100%)
- Tampering heatmap
- Technical report PDF (for legal proof) - unique feature which will help in legal proceedings. (Will find many 3rd party libraries to generate pdf and even we can make it downloadable and send it via email)

2. Secure Reporting Portal

Student can:

- Upload image/video
- Add optional description
- Choose:
 - Anonymous mode
 - Confidential mode

System automatically:

- Encrypts files
- Generates case ID
- Sends notification to college safety cell (or Principal, Director)

3. Evidence Locker (Legal Ready)

System:

- Timestamped hashing (SHA-256)
- Stores digital fingerprint
- Creates downloadable report

This helps in:

- Filing FIR
- Submitting to cybercrime portal
- College disciplinary action

4. Mental Health Support Integration

After detection:

- Provide:
 - Campus counselor contact
 - Women helpline (1091 India)
 - Cybercrime reporting link

Some notes/practical approaches about the Idea:

We cannot freely run our system across multiple social platforms.

We are making a system which allows:

User-Initiated Protection Model (Legal & Practical)

Upload-Based Verification (100% Allowed)

If a student finds:

- A suspicious image
- A fake profile screenshot
- A morphed video

She uploads it voluntarily.

Your system:

- Analyzes the image
- Runs deepfake detection
- Generates report

Public URL Scan (Limited + Ethical)

You can allow:

- User pastes a public URL
- System fetches only that page
- Extracts media
- Runs analysis

Important:

- Only for public content (of public accounts only)
- No automated mass crawling
- Add disclaimer

Reverse Image Search Instead

Instead of scanning Instagram:

Use:

- Image hashing
- Perceptual hashing (pHash)
- Google reverse image API

Workflow:

- Student uploads her original photo
- System generates hash
- Compares with suspicious image
- Detects morphing or reuse

This avoids platform scraping.

Browser Extension (Very Clever & Unique)

Build:

A Chrome extension

When a student is browsing:

- She clicks extension on a suspicious profile
- Extension captures visible image
- Sends to backend for analysis

This way:

- User is manually triggering scan
- No automated scraping
- Fully compliant

Solution 1: Digital Fingerprinting (Proactive Protection)

How It Works:

When a student registers:

1. She uploads her **original photos**
2. System generates:
 - Perceptual hash (pHash)
 - Face embedding vector (using FaceNet / similar)
3. Store these securely

Now if:

- She finds a suspicious image later
- Or uploads a screenshot of a profile

System compares:

Face Embedding Similarity Score
+
Image Structural Similarity
+
Deepfake Detection Score

Even if:

- Cropped
- Filtered
- Slightly morphed

Face embeddings can still match.

- ✓ This works even if image is modified
- ✓ No platform scraping required

Solution 2: User-Triggered Web Scan

Instead of scanning the whole internet:

Let the student:

- Upload suspicious image
- Paste public profile URL

System:

- Extracts visible profile picture
- Compares face embeddings with registered user

This is:

- ✓ Legal (public content only)
- ✓ Triggered manually
- ✓ Deployable

Solution 3: Reverse Image Search API (Smart Integration)

Instead of building your own crawler, integrate:

- Google Reverse Image Search
- Bing Visual Search API

Workflow:

1. Student uploads original photo
2. System calls image search API
3. Returns similar images found online
4. Student reviews suspicious ones

This shifts detection responsibility to search engine infrastructure.

Most Advanced (Optional Bonus Feature)

AI-Based Morph Detection

Even if someone:

- Adds new background
- Alters skin tone
- Slightly reshapes face

You detect using:

- Landmark inconsistency
- GAN artifact detection
- Frequency domain analysis

That means even if:

"It looks real to humans"

Your model detects manipulation artifacts.

Tech Stack (Deployable & Realistic)

Frontend:

- React (Student dashboard)
- Tailwind / Bootstrap

Backend:

- Node.js (Express)
- Python microservice for AI model

AI:

- PyTorch / TensorFlow
- Pretrained deepfake datasets:
 - FaceForensics++
 - DeepFake Detection Challenge Dataset

Storage:

- AWS S3 (encrypted)
- MongoDB (case tracking)

Security:

- JWT authentication
- End-to-end encryption
- File hashing

Future Extensions

- College-wide awareness dashboard - reports of the past, incidents etc.
- Real-time WhatsApp bot for quick reporting - embedded in web or app.
- Auto complaint draft generator
- AI model trained specifically on Indian dataset

Hackathon Ideas - BharatTech

Problem 1:

Title: Face Recognition Surveillance System and Communication Systems for Missing Persons or Items at Simhastha Ujjain for the Police Department

Category: Smart Automation

Description:

Simhastha Ujjain, one of the largest religious gatherings in India, witnesses millions of pilgrims and tourists congregating in a single location. Managing such a vast crowd poses significant challenges for the police department, particularly in ensuring the safety and security of individuals and locating missing persons or items. The deployment of advanced technologies like face recognition surveillance systems and efficient communication systems is critical to address these challenges.

Issues Faced

1. High Volume of Missing Persons and Items:

Overwhelming Number of Reports: During Simhastha, the number of reports for missing persons and items surges, making it difficult for the police to manage and respond promptly.

Identification Challenges: Identifying missing persons in a vast, densely populated area is challenging without advanced technology.

2. Inefficient Manual Processes:

Time-Consuming Identification: Traditional methods of identifying missing persons through manual verification of photos and descriptions are time-consuming and prone to errors.

Delayed Response: The manual process leads to delayed response times, reducing the chances of quickly locating missing persons or items.

3. Communication Barriers:

Lack of Real-Time Updates: Communication between different units and officers is often delayed, affecting coordination and timely response.

Inefficient Information Dissemination: The absence of a centralized system to disseminate information about missing persons or items hampers effective communication and coordination.

4. Security Concerns: Crowd Control: Managing and monitoring large crowds to prevent and respond to incidents is challenging without real-time surveillance and communication.

Potential Threats: The risk of criminal activities increases in such large gatherings, necessitating advanced surveillance for timely threat detection and intervention.

Impact on Operations

1. **Reduced Efficiency:** The inability to quickly and accurately identify and locate missing persons or items results in reduced operational efficiency and effectiveness.
2. **Public Safety Risks:** Delays in responding to reports of missing persons or items pose significant safety risks, especially for vulnerable individuals such as children or the elderly.
3. **Increased Operational Burden:** The police department faces an increased burden due to the high volume of cases and the need for extensive manual effort.
4. **Eroded Public Trust:** Inefficient handling of missing persons or items can erode public trust in the police department's ability to ensure safety and security during large events.

Solution Requirements:

1. Face Recognition Surveillance System:

Real-Time Identification: Deploy face recognition technology to identify and track missing persons in real-time from surveillance footage.

Integration with Existing Systems: Integrate face recognition with existing CCTV infrastructure for seamless operation and monitoring.

Automated Alerts: Implement an automated alert system to notify officers immediately when a match for a missing person or item is detected.

2. Enhanced Communication Systems: Centralized Communication Platform: Establish a centralized platform for real-time communication and information sharing among police units.

Mobile Connectivity: Provide mobile connectivity solutions to ensure that officers on the ground can receive and relay information quickly.

Data Sharing with Public: Set up a system for real-time data sharing with the public, enabling individuals to report missing persons or items and receive updates.

3. Training and Support: Comprehensive Training: Provide training to police officers on using face recognition technology and the new communication systems effectively.

Technical Support: Ensure continuous technical support for the maintenance and troubleshooting of the deployed systems.

4. Privacy and Security Measures:

Data Privacy: Implement robust data privacy measures to protect the personal information of individuals captured by the face recognition system.

Security Protocols: Establish security protocols to safeguard the surveillance and communication systems from unauthorized access and misuse. By addressing these challenges with the implementation of face recognition surveillance and enhanced communication systems, the police department can significantly improve its ability to manage large crowds, ensure public safety, and respond promptly to reports of missing persons or items during Simhasth Ujjain.

Problem 2:**Title:** Hardware Inventory Management in the Police Department**Category:** Smart Resource Conservation

Description: Effective hardware inventory management is crucial for the police department to ensure that all technological assets, such as computers, communication devices, servers, and other equipment, are available, functioning, and up-to-date. Proper inventory management supports the department's operational efficiency, readiness, and ability to respond promptly to various situations. However, the police department currently facing several challenges in managing its hardware inventory, leading to inefficiencies and operational risks.

Issues Faced**1. Lack of Centralized System:**

Decentralized Records: Hardware inventory data is often maintained in a decentralized manner, leading to inconsistencies and difficulties in tracking assets across various locations.

Inefficient Tracking: Without a centralized system, tracking the status, location, and usage of hardware assets becomes cumbersome and prone to errors.

2. Inaccurate Inventory Records:

Outdated Information: Inventory records are frequently outdated due to manual entry errors, infrequent updates, or lack of proper inventory audits.

Missing Assets: The absence of accurate tracking leads to missing or unaccounted-for hardware, which can result in financial losses and operational disruptions.

3. Inefficient Allocation and Utilization:

Resource Misallocation: Inefficient inventory management results in the misallocation of hardware resources, where some units may have surplus equipment while others face shortages.

Underutilization: Proper utilization of hardware assets is hindered, leading to underutilization of available resources and increased operational costs.

4. Maintenance and Lifecycle Management:

Delayed Maintenance: Without accurate tracking, scheduling regular maintenance and timely upgrades for hardware becomes challenging, leading to increased downtime and potential failures.

Lifecycle Tracking: Inadequate tracking of the lifecycle of hardware assets results in delays in replacing obsolete equipment, impacting overall performance and security.

5. Compliance and Security Risks:

Non-Compliance: Inaccurate inventory records can lead to non-compliance with regulatory requirements and internal policies regarding asset management and data protection.

Security Vulnerabilities: Untracked or poorly managed hardware can pose security risks, as outdated or compromised devices may remain in use without detection.

6. Cost Management:

Increased Costs: Inefficiencies in inventory management lead to unnecessary purchases and higher maintenance costs, straining the department's budget.

Budgeting Challenges: Accurate budgeting and financial planning become difficult without precise knowledge of the existing hardware inventory and future needs.

Impact on Operations

1. **Operational Delays:** Inefficiencies in hardware inventory management can cause delays in deploying necessary equipment for critical operations.
2. **Increased Downtime:** Poor maintenance and delayed replacements lead to increased downtime of crucial hardware, affecting the department's overall productivity.
3. **Resource Wastage:** Misallocated underutilized hardware assets contribute to wastage of resources, both in terms of financial and operational capacity.
4. **Security Incidents:** Inadequate management of hardware can result in security breaches, exposing sensitive data and compromising operational integrity. Solution

Requirements

1. **Centralized Inventory Management System:** Integrated Platform: Implement centralized, integrated inventory management system that allows for real-time tracking and updating of all hardware assets across different locations. |ö |ç |

Automated Tracking: Utilize automated tracking technologies, such as barcode scanning or RFID, to ensure accurate and efficient recording of inventory data. 2.

Regular Audits and Updates: |ö |ç | **Routine Audits:** Conduct regular audits of hardware inventory to ensure records are accurate and up-to-date. |ö |ç | **Timely Updates:** Implement protocols for timely updates to the inventory system following acquisitions, disposals, or relocations of hardware assets. 3. **Efficient Allocation and**

Utilization:

Dynamic Allocation: Develop a dynamic allocation system to ensure hardware resources are distributed based on current needs and usage patterns.

Utilization Monitoring: Monitor the utilization rates of hardware assets to identify underutilized resources and optimize their deployment. 4. **Maintenance and**

Lifecycle Management: **Scheduled Maintenance:** Establish a maintenance schedule based on the lifecycle of each hardware asset to prevent unexpected failures and downtime.

Lifecycle Management: Implement a lifecycle management process to track the age and condition of hardware, ensuring timely replacements and upgrades. 5.

Compliance and Security Measures: **Compliance Tracking:** Ensure the inventory management system which includes compliance tracking features to meet regulatory and policy requirements.

Security Protocols: Implement security protocols for tracking and managing hardware assets to mitigate risks associated with lost or compromised equipment.

6. Cost Management and Budgeting:

Cost Analysis: Use inventory data to perform cost analysis and identify

opportunities for cost savings through efficient management and procurement practices.

Budget Forecasting: Utilize accurate inventory records to inform budgeting and financial planning, ensuring adequate funding for future hardware needs. By addressing these challenges through the implementation of a robust hardware inventory management system, the police department can improve operational efficiency, ensure resource availability, and enhance overall effectiveness in its mission to maintain public safety and order.

Problem 3:

Title: Development and Implementation of Face Recognition Technology in the Police Department

Category: Smart Automation

Description: The police department aims to enhance its operational capabilities through the development and implementation of face recognition technology (FRT). This advanced technology can significantly aid in identifying suspects, locating missing persons, and maintaining public safety. However, the deployment of FRT involves complex challenges that need to be addressed to ensure its effectiveness, reliability, and ethical use.

Issues Faced

1. **Technical Challenges: Accuracy and Reliability:** Developing FRT that provides consistent and accurate results across various conditions (lighting, angles, occlusions) and diverse populations technically challenging. False positives and negatives can undermine the credibility and utility of the system.

Integration with Existing Systems: The new technology must seamlessly integrate with existing police databases and IT infrastructure, which can be complex and resource-intensive.

2. Data Quality and Management:

Database Quality: The effectiveness of FRT is heavily dependent on the quality and completeness of the reference database. Inaccurate or outdated data can lead to misidentification.

Data Volume and Speed: Handling large volumes of data in real-time requires robust data processing capabilities and efficient algorithms.

3. **Ethical and Privacy Concerns:****Privacy Invasion:** FRT has the potential to infringe on individual's privacy rights, leading to public distrust and resistance. There is a need to balance surveillance benefits with privacy protections.

Bias and Discrimination: FRT systems can exhibit bias against certain demographic groups, leading to unfair treatment and potential civil rights violations. Ensuring fairness and mitigating bias are crucial

4. **Legal and Regulatory Issues:****Lack of Clear Regulations:** The rapid advancement of FRT has outpaced the development of comprehensive legal frameworks. Clear regulations are needed to govern its use and protect citizens' rights

Compliance with Existing Laws: The development and use of FRT must comply with existing privacy and data protection laws, which may vary across jurisdictions.

5. Operational and Training Needs: User Training. Effective use of FRT requires training law enforcement personnel in its operation, interpretation of results, and ethical considerations.

Maintenance and Support: Ongoing technical support and system maintenance are necessary to ensure the technology remains effective and up-to-date.

6. Public Perception and Acceptance: Transparency and Accountability Building public trust requires transparency in how FRT is used and accountability mechanisms to address misuse. **Community Engagement:** Engaging with the community to educate them about the benefits and safeguards of FRT can help mitigate concerns and gain public support.

Impact on Operations

1. Enhanced Identification Capabilities: When effectively implemented, FRT can significantly enhance the ability to identify and apprehend suspects, improving public safety.

2. Resource Allocation: Development and implementation require substantial financial and human resources, which must be justified by the benefits and outcomes.

3. Legal and Ethical Implications: Misuse or ineffective use of FRT can lead to legal challenges, ethical dilemmas, and public backlash.

4. Operational Efficiency: Proper integration and use of FRT can improve operational efficiency, reducing the time and effort required for manual identification processes.

Solution Requirements

1. Development of Robust Algorithms: Focus on creating advanced algorithms that ensure high accuracy, minimize bias, and handle diverse conditions effectively.

2. Comprehensive Legal Framework: Advocate for and contribute to the development of clear regulations that govern the ethical use of FRT.

3. Privacy and Security Measures: Implement strong data protection protocols to safeguard biometric data and ensure compliance with privacy laws.

4. Training Programs: Develop comprehensive training programs for law enforcement personnel to ensure they can use FRT effectively and ethically.

5. Public Engagement Initiatives: Establish initiatives to engage with the community, address concerns, and build public trust in the use of FRT.

6. Continuous Monitoring and Evaluation: Set up mechanisms for continuous monitoring, evaluation, and improvement of the FRT system to ensure it meets

operational needs and ethical standards. By addressing these challenges, the police department can develop and implement face recognition technology that enhances law enforcement capabilities while maintaining public trust, protecting privacy, and ensuring ethical use.

Problem 4

Title: Fuzzy Name conversion of Hindi Names in Police Records

Category: Miscellaneous

Description: The police department handles a vast array of data including the names of individuals involved in various activities such as reporting incidents, being suspects, witnesses, or victims. A significant challenge arises with the accurate recording and retrieval of these names, especially when dealing with Hindi names, which can be represented in multiple ways due to variations in transliteration, spelling, and pronunciation.

Issues Faced

1. **Inconsistent Transliteration:** Hindi names often have multiple valid English transliterations. For example, the name Suresh in Hindi could be written as "Suresh" or "Sursh". This inconsistency creates difficulties in searching and matching records accurately.
2. **Spelling Variations:** Minor spelling differences, such as "Suresh" vs. "Sursh", can lead to significant discrepancies in the database, resulting in incomplete or incorrect search results.
3. **Phonetic Similarity:** Different names that sound similar phonetically but have different spellings can be a challenge. For instance, "Kumar" and "Kumaar" may refer to the same individual but are treated as distinct entries in the system.
4. **Data Entry Errors:** Human errors during data entry can introduce further variations, such as typos or incorrect transliterations, making it harder to maintain accurate and consistent records.
5. **Multiple Scripts:** The use of both Devanagari (Hindi script) and Roman script adds another layer of complexity. A name recorded in Hindi may need to be searched in its English transliterated form, and vice versa.
6. **Search Efficiency:** Traditional search algorithms may not effectively handle these variations, leading to inefficiencies in retrieving relevant records during investigations or administrative processes.

Impact on Operations

1. **Delayed Investigations:** Inaccurate name matching can lead to delays in criminal investigations as officers struggle to retrieve all relevant records associated with a particular individual.
2. **Resource Wastage:** Additional resources are required to manually cross-check and verify records that the system fails to match automatically, increasing operational costs and workload.

3. Legal Implications: Inaccurate or incomplete records can have serious legal implications, potentially affecting the outcomes of cases and leading to miscarriages of justice.

4. Public Trust: Inefficiencies and inaccuracies in handling sensitive information can erode public trust in the police department's ability to manage and process data effectively.

Solution Requirements

1. Fuzzy Matching Algorithms: Implement advanced fuzzy matching algorithms capable of handling variations in spelling, transliteration, and phonetic similarities.

2. Standardized Transliteration: Establish and enforce standardized rules for the transliteration of Hindi names into English inconsistencies. to minimize

3. Phonetic Search Capability: Develop search capabilities that consider phonetic similarities to improve the accuracy of name matching.

4. Error Correction Mechanisms: Integrate error correction mechanisms to identify and rectify common data entry errors.

5. Script Interoperability: Ensure seamless interoperability between Devanagari and Roman scripts to facilitate accurate searches across different scripts.

6. Training and Guidelines: Provide training to personnel on best practices for name entry and establish clear guidelines to minimize errors and inconsistencies. By addressing these issues, the police department can significantly enhance the accuracy and efficiency of its name matching processes, leading to more effective law enforcement operations and improved public trust.

Problem 5:

Title: A Software, AI App to provide legal information ie. Case laws various important landmark judgement on important case legal matters.

Category: Smart Automation

Description:

Background: Due to unavailability of legal experts in Police stations, whenever a complainant comes to us with his/her complaint to get a First Information Report written sometimes we make mistakes in writing all the sections/acts due to which problems arise in the investigation later on.

Description: Whenever a complainant comes to the police station with his/her complaint to get the First Information Report written, the investigating officer sometimes due to ignorance or lack of experience is not able to follow the provisions and the First Information Report remains deficient. Due to which, later on, problems arise in the investigation and justice gets affected, as legal experts are not posted at police stations. If there will be any such software or app which can use Artificial Intelligence (AI) and immediately provide us the information regarding appropriate and correct sections and acts, as per the incident of the complainant. So there will be no errors regarding the sections and acts under the First Information Report and the investigation and justice will not be affected and will help in investigation. Expected Solution: We want a software (app) in which as soon

as we write/speak the details of the incident, information about appropriate sections and acts is immediately available to us and which is reliable

Problem 6

Title: Domain hosting details

Category: Cybersecurity

Description: On searching many suspicious websites on whois, their hosting appears to be on cloud flare, but after sending mail to cloud flare, say that the said website uses one of their services like SSL certificate etc. and they do not have the hosting of the website, due to which correct information about the suspicious website cannot be obtained. Challenges: Correct information from suspicious websites is not available. Expected Solution: A mechanism should be developed that can provide accurate information about suspicious websites along with their actual hosting.

Problem 7

Title: Conversion of 2D Blueprints into 3D Model

Category: Smart Automation

Description: 1. National Security Guard (NSG) operates inside buildings, built up areas and other infrastructures like metro stations, schools, etc in case of any terror. Since the operations are of National security significance, better understanding of the target layout is of paramount importance.

3-Dimensional walk through of the buildings helps in the efficient understanding of the target layout and also enables effective briefing of the troops by the Commander for operating inside such targets.

2. Presently, the building authorities provide 2-Dimensional layout of the building in the form of blue prints/ floor plan. Assimilation of the situation and appreciating the potent threat with the help of 2-Dimensional layout turns out to be a challenging task. In such a scenario, a 3-Dimensional walk-through of the building helps significantly in increasing the efficiency of the ground forces while operating.

3. The software which needs to be developed should take the 2-Dimensional blueprint layout as input and also should take various parameters like length, breadth, height of the building, layout of staircase, entry and exit, etc as additional inputs. The software should then export them to a file and create a full fledged 3-Dimensional walkthrough model of the building. The software must also include the facility for integration of exported model to the offline Google maps, Satellite pictures (if any available).

4. Since the target includes buildings which are secretive in nature, the software is required to be operable fully in standalone mode, without any internet connectivity required in future. For the initial installation phase, the system can be connected to the internet to facilitate fetching up of Maps, satellite imagery from open source, etc required for the software functioning.

5. This software would also reduce the time taken during the initial briefing of the troops prior to the launch of any operation. It further increases the efficiency of the operations, thus resulting in early success. 6. The conversion of 2-Dimensional blueprints to 3-Dimensional models must address following problems. (a) Software

should take 2-Dimensional blueprint as an input and produce 3- Dimensional walk through as output. (b) Provisions to input parameters like length, breadth, height of the building and other details should be included. (c) Minimum supervision from the user end should be needed to finalise the conversion. (d) Offline integration with Google maps, satellite imagery to get enhanced output covering the nearby geographical areas as well.

Problem 8

Title: AI-Powered Delivery Post Office Identification System The wide, evolving delivery network of the Post Office makes it difficult for customers to write the correct pin code on the postal items for delivery. The Post Office also merges Pincodes together to mechanise and streamline delivery in the emerging volume and mix of mail handled at different stages, including the point of delivery. An intelligent solution, powered by AI is needed to meet the dynamic design of the supply chain both for customers as well as operators within India Post.

Category: Transportation & Logistics

Description:

Background: The Department of Post in India boasts the world's largest postal network, with over 165,000 post offices. Through nearly 19,000 pin codes across the country the postal network manages the delivery of postal items across its network. One common issue is the mismatch between the specified Delivery Post Office name and the area PIN code on mail items, leading to delays and customer dissatisfaction. The delivery network design keeps on evolving as Nodal Delivery Centres and Delivery Hubs are introduced to bring in more efficiency in operations. From the perspective of the customer, the challenge is two fold: (a) to know the correct Pin Code, which is itself a challenge; (b) the dynamic nature of operations inside with merged pin codes etc for all or certain categories of mail is also a challenge. Though the customer does not need to bother about (b), the internal system of India Post needs to take care of both issues (a) and (b) for efficient and correct delivery at all times. Description: Postal services are essential for various government, financial, educational, corporate, and public entities, both in urban and rural areas. Despite the widespread use of postal services, inaccurate addressing persists, hampering efficient delivery. Each mail item must clearly indicate the recipient's Delivery Post Office name and area Pin Code. However, inconsistencies in this information often result in misrouted mail, with approximately 5% of articles handled daily at distribution hubs containing incorrect Pin Codes. Even if correct Pincodes are provided as per information available with the customer, there is a mismatch in the internal operations as the constant upgradation of delivery network with Nodal Delivery Centres with merged pincodes etc makes it too dynamic for internal operators to keep pace with the changes. This leads to mistakes, delays and wasteful movement of mail pieces in the system. Expected Solution: An AI-based scanning solution is proposed to identify the correct delivery location for mail items with unclear or mismatched addresses. This system will utilize advanced machine learning algorithms to analyze and interpret address information, determining the appropriate Delivery Post Office based on available data. By automating the identification process, this solution aims to improve last-mile delivery efficiency, reducing delays and enhancing customer satisfaction. Having taken care of the first problem, the system is also expected to align with internal operational databases for clubbing and mapping of different

pincodes for delivery purpose and thereby suggest the right destination for dispatch and delivery of the articles.

Problem 9:

Title: Farmers Disease Diagnostic/Reporting Portal - Mobile Portal AI Based

Category: Agriculture, FoodTech & Rural Development

Description:

Background- Agricultural productivity and food security are heavily dependent on the health of crops and livestock. Farmers, especially in remote or resource-limited areas, often face challenges in diagnosing and reporting diseases that affect their livestock and crops. Early diagnosis and timely reporting are crucial for effective disease management and prevention of widespread outbreaks.

Description - A mobile portal powered by Artificial Intelligence (AI) for disease diagnosis and reporting can revolutionize how farmers manage animal and plant health. This portal enables farmers to use their smartphones to access diagnostic tools, report symptoms, and receive actionable advice. AI algorithms can analyze reported data to provide accurate diagnoses and suggest appropriate treatments or interventions. This technology can empower farmers with the knowledge and tools to manage diseases more effectively, ensuring the health and productivity of their farms. Integrate AI-based software with existing NDLM to allow farmers to submit images and descriptions of disease signs and symptoms. The AI will generate suspected disease/condition reports with preventive measures and send alerts to veterinarians for appropriate action (AI-based software, Mobile application).

Expected Outcomes . Enhanced Disease Diagnosis: . Timely Reporting and Intervention: . Increased Access to Expert Knowledge: . Improved Farm Productivity. Data Collection and Analysis: Cost-Effective DiseaseManagement: Empowerment and Education of Farmers: Integration with Existing Surveillance Systems . Community Engagement and Support: Sustainable Agriculture Practices

1. Smart Tourist Safety Monitoring & Incident Response System

PS ID: SIH25002

Organization: Ministry of Development of North Eastern Region

Theme: Travel & Tourism

Problem

Tourists traveling in remote regions (mountains, forests, rural areas) often face:

- getting lost
- accidents
- crime
- slow emergency response

The government wants a **digital system to monitor tourist safety and respond to incidents quickly.**

Why This Is a Strong Hackathon Choice

- **Huge real-world impact**
- **Pure software solution**
- **Easy to add AI + geospatial + blockchain features**
- **Judges like tourism + safety projects**

Unique Features You Can Add (to stand out)

1. AI Risk Prediction

AI predicts **danger zones** using:

- crime data
- weather
- terrain
- accident history

Shows **risk heatmap** for tourists.

2. Smart SOS System (Already built)

One-tap emergency alert:

- sends **location + live audio/video**
- alerts **nearest police + rescue team**

3. Geo-Fence Safety Alerts

If tourist enters:

- restricted forest
- landslide zone
- dangerous terrain

The app sends **automatic warning**.

4. Offline Emergency Mode (Already implemented)

In remote areas:

- emergency messages sent via **SMS fallback**
- mesh networking between phones.

5. Blockchain Tourist Identity

Secure digital tourist ID:

- prevents fraud
- helps identify missing tourists.

1. Conversion of 2D Blueprints into 3D Model (Problem 7)

2. AI App for FIR Legal Section Recommendation (Problem 5)